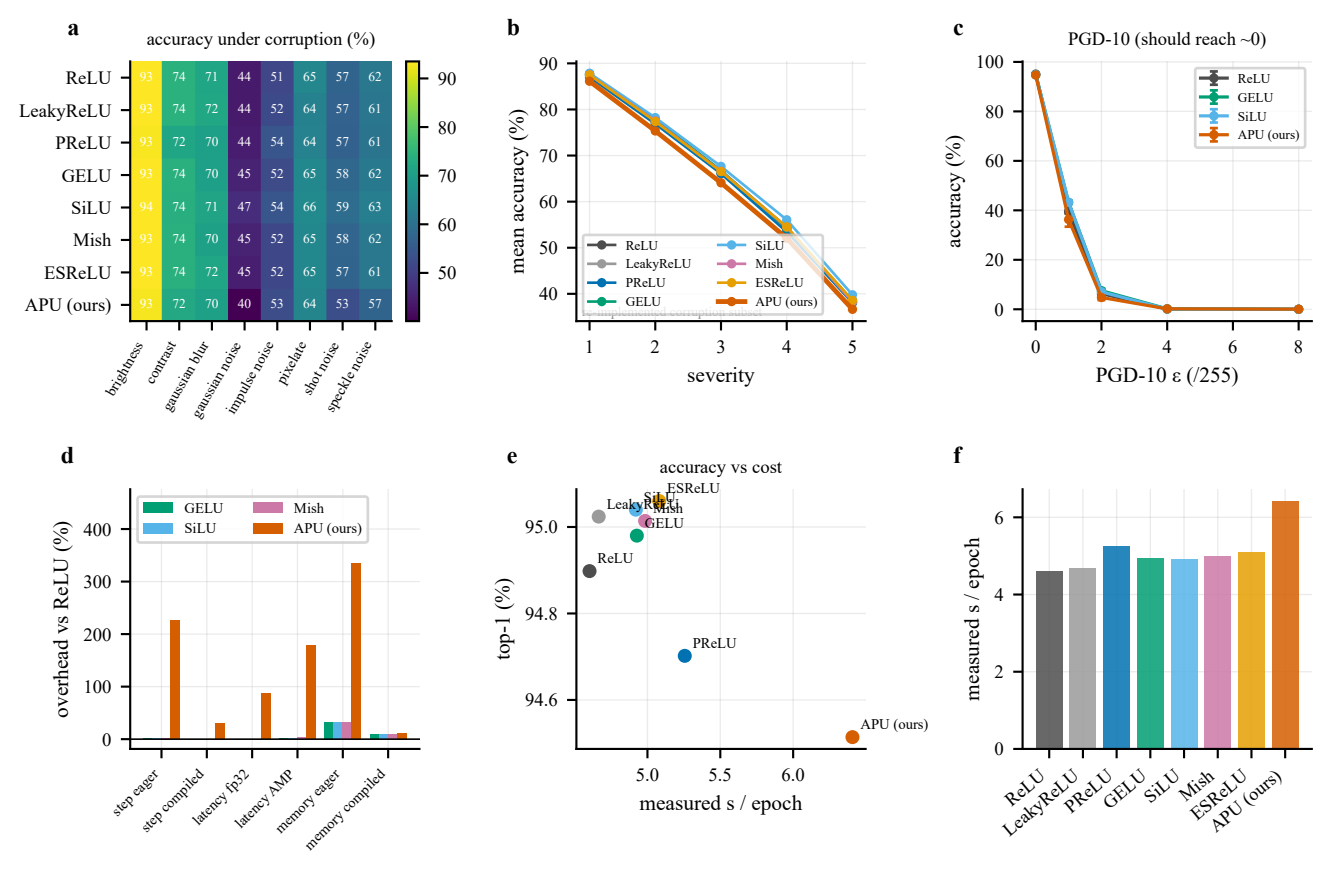

a
accuracy under corruption (%)
ReLU 93 74 71 44 51 65 57 62
LeakyReLU 93 74 72 44 52 64 57 61
PReLU 93 72 70 44 54 64 57 61
GELU 93 74 70 45 52 65 58 62
SiLU 94 74 71 47 54 66 59 63
Mish 93 74 70 45 52 65 58 62
ESReLU 93 74 72 45 52 65 57 61
APU (ours) 93 72 70 40 53 64 53 57
90
80
70
60
50
brightness
contrast
gaussian blur
gaussian noise
impulse noise
pixelate
shot noise
speckle noise
b
mean accuracy (%)
severity
ReLU
LeakyReLU
PReLU
GELU
SiLU
Mish
ESReLU
APU (ours)
c
PGD-10 (should reach ~0)
accuracy (%)
PGD-10 ε (/255)
ReLU
GELU
SiLU
APU (ours)
d
overhead vs ReLU (%)
GELU
SiLU
Mish
APU (ours)
step eager
step compiled
latency fp32
latency AMP
memory eager
memory compiled
e
accuracy vs cost
top-1 (%)
measured s / epoch
ReLU
LeakyReLU
PReLU
GELU
SiLU
Mish
ESReLU
APU (ours)
f
measured s / epoch
ReLU
LeakyReLU
PReLU
GELU
SiLU
Mish
ESReLU
APU (ours)